MILITARY SPECIFICATION

TEST SET, RADIO AN/ARM-32

This specification has been approved by the
Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered in this specification shall be suitable for bench testing of Radio Receiving Sets AN/ARN-41, AN/ARN-41A, AN/ARN-41B, and AN/ARN-54 in conjunction with standard test equipment. The equipment shall have provisions for securing and mechanically connecting those elements of the receiving set which control the function and tuning of the equipment being tested.

1.2 Classification - Radio Test Set AN/ARM-32 shall be of one type and shall consist of the following:

Units	Type Designation	Max. Allow. Size (Inches) W	H	L/D	Max. Allow. Wt. Lbs.	Appl. Para.
Test Set, Radio	TS-1062/ARM-32	12-1/2	14-1/2	20-1/2	18.0	3.3.1
Cover, Test Set	CW-440/ARM-32	12-1/2	14-1/2	1-3/8	3.0	3.3.2
Cable Assembly, Radio Frequency	CG-1568/ARM-32			336	2.0	3.3.3
Cable Assembly, Radio Frequency	CG-1569/W(6' 6")			78	1.0	3.3.4
Cable Assembly, Special Purpose, Electrical, Branched	CX-4140/ARM-32			30	1.75	3.3.5
Cable Assembly, Special Purpose, Electrical Branched	CX-4141/ARM-32			60	2.25	3.3.6
Cable Assembly, Special Purpose, Electrical	CX-4142/ARM-32			46	.75	3.3.7
Shaft Assembly, Flexible	MX-2257/ARM-32				1.0	3.3.8
Mounting	MT-1890/ARM-32				1.5	3.3.9

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

Source: https://assist.dla.mil -- Downloaded: 20260-19T09:06Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

Military

MIL-T-945	Test Equipment, For Use with Electronic Equipment; General Specification
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-E-5422	Environmental Testing; Aircraft Electronic Equipment
MIL-A-8625	Anodic-Coatings for Aluminum and Aluminum Alloys
MIL-E-15090	Enamel, Equipment, Light Gray
MIL-E-15354	Electrical Test and Maintenance Tables, Table Tops, and Back Panel and Shelf Assembly
MIL-H-15362	Test Bench Harness for Electronic Equipment; General Specification
MIL-P-17555	Preparation for Delivery of Electronic Equipment: Accessories, Auxiliary Equipment, and Associated Repair (Maintenance) Parts
MIL-T-18303	Test Procedures, Preproduction and Inspection; Format for
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment

Bureau of Aeronautics

SAR-300	Engineering Change Proposals, Electronic Material Changes and Electronic Material Bulletins; Preparation of

DRAWINGS

Bureau of Ships

S9101-212410	Work Bench, Aviation, Electronic Shop for

2.2 Availability of Documents:

(1) Give both the title and number when requesting specifications, standards, drawings, and publications. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot (AD), Philadelphia 11, Pennsylvania.

(3) Copies of applicable drawings may be obtained upon application to the Bureau of Aeronautics, Navy Department, Washington 25, D.C. Attention: Avionics Division.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:06Z
Check the source to verify that this is the current version before use.

2.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be approved in writing by the Bureau of Aeronautics.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following factors shall govern unless otherwise specified.

(1) Parts and materials as approved by Specification MIL-E-5400 shall be given first consideration.

(2) When the characteristics of standard MIL parts are such that their use will not fulfill the requirements because of size, weight, performance or other reasons, material and parts shall be used which most nearly meet or exceed the requirements of the respective standard parts and material specifications.

(3) When a non-standard part is used, the meeting of general material requirements such as non-inflammability, fungus resistance, non-toxicity, etc., and the meeting of environmental conditions as required by Specification MIL-H-15362 shall be of prime consideration.

3.1.1 Non-Standard Part and Material Approval - Approval for the use of non-standard parts and materials shall be obtained as outlined in Specification MIL-E-5400 except as follows:

(1) Mechanical parts shall be submitted as non-standard parts when the material requirements of Specification MIL-E-5400 are not met.

(2) Non-standard parts of an electrical nature, as defined in Specification MIL-E-5400 must be approved before the preproduction model of the equipment is submitted for tests. The request for approval must be forwarded by the contractor before the part is purchased or fabricated. In addition to or in lieu of the Non-Standard Part Data Sheet, the contractor is requested to submit itemized lists of the parts, their sources, and the contractor's drawing or purchase specification covering the parts.

3.1.2 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives to be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance, or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-H-15362, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the Bureau of Aeronautics for consideration. Each request should be accompanied by complete supporting information.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:06Z
Check the source to verify that this is the current version before use.

3.2 Design and Construction:

3.2.1 General - This equipment shall fulfill all applicable design and construction requirements of Specification MIL-H-15362 unless otherwise specified herein.

3.2.2 Service Bench - The equipment shall be suitable for operation on any service bench described in Specification MIL-E-15354 or Bureau of Ships Drawing S9101-212410 and shall provide the required clamps for mounting the controlled units of the radio receiving set being tested to the bench.

3.2.3 Size and Weight - The equipment shall not exceed 32 pounds in weight including cables nor 12-1/2 inches wide, 15-1/2 inches high by 21-1/2 inches deep in size including rubber feet on the case.

3.2.4 Standard Conditions - The following conditions shall be used as a basis to establish normal equipment performance and for making laboratory bench tests using the equipment:

Temperature	Room Ambient 25°C (+15°C, -5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	27.5 ± 0.5 V DC

3.2.5 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions or reasonable combination of service conditions specified in Specification MIL-H-15362 except that the following requirements shall apply:

3.2.5.1 Equipment Non-Operative

(1) Temperature	-60°C through +85°C
(2) Humidity	in excess of 95%

3.2.6 Operational Stability - Performance of the equipment shall not in any way degenerate during long periods of continuous or intermittent operation.

3.2.7 Nameplates and Nomenclature - Nameplates and nomenclature shall be in accordance with the requirements of Specification MIL-N-18307.

3.2.8 Panel Markings - Identification of operating controls, receptacles and all other markings shall be in accordance with the requirements of Specification MIL-T-945.

3.2.9 Moisture and Fungus Treatment - The equipment shall be fungus proofed by selection of parts and materials as non-nutrient for fungus or the parts and materials shall be treated prior to their use in the equipment so that overall spraying of the equipment is not necessary.

3.2.10 Overload Protection - The equipment shall be protected by a fuse in the 27.5 V DC primary power line.

3.2.11 Front Panel Layout - All operational controls, test jacks, and indicators shall be located on the front panel of Radio Test Set TS-1062/ARM-32. Prior to fabrication of the first equipment, the contractor shall submit a drawing of the proposed panel layout, including identifying titles, to the Bureau of Aeronautics for review and approval.

3.2.12 Controls and Adjustments - All controls, connectors, and test jacks shall be designed, located and labeled to provide for ease of operation and maintenance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:06Z
Check the source to verify that this is the current version before use.

3.2.13 Cables and Connectors - Except as specified under the detail requirements below, cables and connectors shall be in accordance with the requirements of Specification MIL-E-5400. Cable assemblies shall be marked in accordance with the requirements of Specification MIL-E-5400.

3.2.14 Degree of Enclosure - With the covers fastened, the equipment shall be spray-tight and reasonably watertight.

3.3 Detail Requirements -

3.3.1 Radio Test Set TS-1062/ARM-32 - The test set shall meet the following requirements:

3.3.1.1 Function - The equipment shall provide electrical and mechanical connections of the units of the receiving set to each other, to the power source, and to the required accessory test equipment. In addition, the equipment shall provide a screened compartment of known attenuation factor within which the receiving set loop antenna assembly may be conveniently mounted and having the required connections for electrically rotating the loop antenna and observing its position. The equipment shall be suitable for testing Radio Receiving Sets AN/ARN-41, AN/ARN-41A, AN/ARN-41B, and AN/ARN-54 which consist of the following units:

Radio Receiving Sets AN/ARN-41, AN/ARN-54

Radio Receiver

Electronic Control Amplifier

Antenna

Radio Receiving Sets AN/ARN-41A, AN/ARN-41B

Radio Receiver

Electronic Control Amplifier

Radio Set Control

Antenna

3.3.1.2 Form Factor - The dimensions of the test set shall not exceed 12-1/2 inches wide, 14-1/2 inches high, and 20-1/2 inches deep.

3.3.1.3 Contents - The test set shall provide the necessary circuitry to control and facilitate measurements of voltage and current to the radio receiving set being tested, and shall include the following facilities, indicators and functions:

(1) A receptacle for connecting the loop antenna assembly power cable.

(2) A receptacle providing for connection of the radio receiver and electronic control amplifier in such a way that their primary power is supplied from the test set and various voltages delivered from the electronic control amplifier to the radio receiver are tapped off into test points and a meter on the test set.

(3) A receptacle for connecting the input from a signal generator.

(4) A receptacle connecting the sense antenna input receptacle of the receiving set to the test set.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T09:06Z
Check the source to verify that this is the current version before use.

(5) A plug and a suitable mounting hole for installing a Direction Indicator ID-90, or equivalent, in the front panel of the test set.

(6) A receptacle for connecting a second direction indicator in parallel with that used in the test set.

(7) Two telephone type jacks, one connected to the 600 ohm output of the receiving set, the other connected to the 3.2 ohm output.

(8) Banana type test receptacles so connected as to enable the following receiving set characteristics to be checked:

(a) ADF output voltage

(b) 115 cps reference voltage

(c) Audio output level

(d) Radio Receiver B+ voltage input

(e) Primary power voltage (two receptacles with a removable link to enable current to be measured)

(9) A 0-30 V AC scale meter provided with a selector switch to measure the adf output on the 115 cps reference voltage of the receiving set.

(10) A push type momentary contact switch which applies the 115 cps reference voltage (shifted 90° in phase) to the varied phase winding of the loop antenna drive motor.

(11) 27.5 V DC switch and fuse.

(12) A two position toggle switch selecting r-f energy from the signal generator to be fed through the coupling network directly to the receiving set sense antenna input, or to be fed to the divided network causing the input signal to be shared between the sense antenna input receptacle and the radiating rod in the screened chamber.

(13) A two position toggle switch to remove the position signal from the varied phase winding of the loop motor.

(14) Instruction plates showing the equipment's attenuation factor, intercabling diagram and summary operation instruction for final testing of the receiving set.

(15) Storage space for all accessories supplied with the test set.

3.3.1.3.1 Front Panel Controls, Connectors and Indicators - All controls, connectors, and indicators shall be mounted on the front panel which shall be attached in such a way as to be easily removeable for servicing. The RG-58A/U cable connecting the signal generator input divider network with the radiating rod shall terminate in a PL-259 plug using a UG-175/U adapter. The plug shall mate with an SO-239 receptacle mounted on the bulkhead separating the screened compartment from the remainder of the test set.

3.3.1.3.2 Cover Plate - In the event a direction indicator is not mounted in the equipment at the time of manufacture, a cover plate shall be attached to cover the direction indicator mounting hole. The cover plate shall be easily removable with ordinary hand tools and shall be plainly marked to indicate its purpose and that it should be thrown away upon mounting a direction indicator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:06Z
Check the source to verify that this is the current version before use.

3.3.1.3.3 Screened Compartment - The screened compartment shall measure 12 inches by 12 inches by 14 inches. The radiating rod shall consist of 12 inches of 1/8 inch diameter brass rod and shall terminate in one end in an SO-239 receptacle and at the other end in an insulator. The rod shall be firmly affixed in such a way as not to be repositioned by the components stored in the screened compartment while the equipment is in transit. A mounting hole in the top of the screened compartment shall be provided for Antenna AT-556A/ARN-41 or AT-556B/ARN-41. Speed nuts shall be provided so that the antenna may be firmly secured in place using all four of its mounting holes. The attenuation factor of the screen compartment shall be as stated in 3.3.1.3(14).

3.3.1.3.4 Antenna Cover - A cover shall be provided for the antenna mounting opening which is so gasketed as to be spray tight and reasonable watertight. The cover shall be attached to the test set by a chain.

3.3.1.3.5 Stowage - The stowage compartment in the front of the test set, above the front panel, shall be designed to hold Cable Assemblies CX-4140/ARM-32, CX-4141/ARM-32, and CG-1569/U as well as two each of Mounting MT-1890/ARM-32. The cover of the compartment will be piano-hinged to swing downward and be fastened when closed by a Camlock fastener.

3.3.1.3.6 The test set shall be finished in light grey enamel in accordance with Specification MIL-E-15090.

3.3.2 Cover, Test Set CW-440/ARM-32 - The cover shall completely enclose the front of the test set so as to make the equipment spray tight and reasonably watertight. Two suitcase fasteners, one at each side, shall clamp the cover in place when closed. The cover shall be completely removable and shall not be attached to the test set in any way except by the suitcase fasteners.

3.3.3 Cable Assembly, Radio Frequency CG-1568/ARM-32 - This cable shall be 27.5 feet in length excluding connectors. The twin coaxial cable shall be Belden Manufacturing Co. No. 8227, or an approved equivalent. It shall be terminated at one end by an AN3106B-10SL-4S right-angle connector. This cable shall be used for connecting the loop receptacle on the test set with the loop rf input receptacle on the receiving set.

3.3.4 Cable Assembly, Radio Frequency CG-1569/U (6 feet - 6 inches) - This cable shall be 73.5 inches in length excluding connectors. It shall consist of RG-114/U cable with PL-259 connectors at either end. This cable shall be used to interconnect the SENSE ANT receptacle on the test set with the sense antenna input receptacle on the receiving set.

3.3.5 Cable Assembly, Special Purpose, Electrical, Branched CX-4140/ARM-32 - This cable shall have four branches extending at right angles from the center stem. The center stem shall be 10 inches, two of the branches 6 inches long, and the remaining two branches 8 inches long. The cable shall consist of 24 wires with suitable outer covering. Twenty-two of the wires shall be 20 gauge and the remaining two, 18 gauge. The following connectors shall terminate the cable branches: one AN3107A-20-16S, one AN3107A-27P, one AN3107A-20-27S, one Lear No. 710907, or an approved equivalent. The r-f connections at connector 710907 shall terminate in one AN3102A-10SL-4P and one SO-239 connector. The cable shall be suitable for interconnecting the components of Radio Receiving Sets AN/ARN-41 and AN/ARN-54 with the test set.

3.3.6 Cable Assembly, Special Purpose, Electrical, Branched CX-4141/ARN-32 - This cable shall have four branches extending at right angles from its center stem and a fifth branch axially located to the center stem. The center stem shall be 10 inches long, the axial branch 24 inches long, two of the right angle branches 18 inches long, and the remaining two branches 2 inches long. The cable shall consist of 32 wires, with suitable outer covering. Twenty-nine of the wires shall be 20 gauge and the remaining two 18 gauge. The following connectors shall terminate the cable branches: One AN3107A-20-29S, one AN3107A-20-16S, one AN3107A-20-27P, one AN3107A-20-27S, and one Lear No. 710907, or an approved equivalent. The rf connections at connector 710907 in one AN3102A-10SL-4P and one SO-239 connector. The cable shall be suitable for interconnecting the components of Radio Receiving Sets AN/ARN-41A, and AN/ARN-41B with the test set.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:06Z
Check the source to verify that this is the current version before use.

3.3.7 Cable Assembly, Special Purpose, Electrical CX-4142/ARM-32 - This cable shall be 42 inches in length, excluding connectors, and shall consist of five 20 gauge wires suitably covered. The cable shall terminate with an AN3108B-14S-6S connector at one end and an AN3107A-14S-6S connector at the other end. The cable shall connect the test set with Antenna AT-556A/ARN-41 or Antenna AT-556B/ARN-41.

3.3.8 Shaft Assembly, Flexible MX-2257/ARM-32 - The flexible shaft assembly shall be 47 inches long, excluding connector fittings, and shall be covered by vinyl sleeving. The shaft shall be suitable for connecting the radio set control with the radio receiver while the radio receiving set is being tested.

3.3.9 Mounting MT-1890/ARM-32 - There shall be two of these mountings supplied with each radio test set. The mountings shall be of the screw clamp type with the thumbscrew captivated in such a way as to prevent inadvertent separation of the clamp into its component parts. The clamp shall be large enough to accomodate any of the radio receivers supplied with the referenced receiving sets and shall be drilled on the proper centers to enable the clamp to be bolted securely to a standard workbench as defined in Specification MIL-E-15354 and Bureau of Ships Drawing 59101-212410. The clamps shall be lacquer finished in accordance with Specification MIL-A-8625.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:06Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine conformance with all the applicable requirements:

(1) Preproduction Tests

(2) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.4. Data obtained by the contractor in conducting these tests shall be submitted to the Bureau of Aeronautics for review and approval prior to shipping the equipment to the specified destination for service approval tests. The Government Inspector and the Bureau of Aeronautics shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory satisfactory to the Bureau of Aeronautics.

4.2.2 Service Approval Tests - The service approval model shall consist of one equipment representative of the production equipment to be supplied under the contract. This may be the same equipment that was subjected to the contractor's demonstration tests. This equipment shall be delivered to the destination specified by the Bureau of Aeronautics. When the tests of 4.2.1 are supervised, the Bureau of Aeronautics may, when conducting service approval tests, omit all tests which will duplicate tests previously conducted by the contractor; however, in conducting service approval tests, the Bureau of Aeronautics reserves the right to repeat any test previously conducted when deemed necessary.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-T-945. The design and test data shall include the data obtained by the contractor in testing to Specification MIL-E-5422. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-E-5422

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T09:06Z
Check the source to verify that this is the current version before use.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the Bureau of Aeronautics shall engage the service of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Reliability Test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements of Specification MIL-T-945 have been met.

4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.3 Reliability Test - Each equipment shall be operated for a period of 6 hours without failure, under the conditions specified herein. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodially through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:06Z
Check the source to verify that this is the current version before use.

4.3.2 Sampling Tests - Sampling tests shall be conducted on complete equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests.

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
11 to 100	2
101 to 200	1
Over 200	1 for each additional 200 or fraction thereof

4.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions and nominal input voltages, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Repeat the test of (1) above using the highest and lowest specified input voltages and frequencies.

(3) Reliability test specified in 4.3.1.3, except that the test duration shall be 30 hours.

4.3.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling test.

4.3.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the Bureau of Aeronautics.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory quality control.

4.3.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the Bureau of Aeronautics. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the Bureau of Aeronautics for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only such tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

Source: https://assist.dla.mil -- Downloaded: 20261O-19T09:06Z
Check the source to verify that this is the current version before use.

(2) Vibration Test (90 minutes cycling - in each of 3 planes)

(3) Humidity Tests

4.3.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the Bureau of Aeronautics for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Test Procedures - The procedures used for conducting preproduction tests, and acceptance tests shall be prepared by the contractor and submitted to the Bureau of Aeronautics for review and approval. The right is reserved by the Bureau of Aeronautics or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the Bureau of Aeronautics will furnish procedures to be used for testing.

4.5 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply,to the test of his knowledge and belief, with all applicable requirements.

4.6 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the Bureau of Aeronautics.

4.7 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the Bureau of Aeronautics. To obtain Bureau of Aeronautics approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a Bureau of Aeronautics' recommended change or a contractor's proposed change. When required, the contractor shall also prepare an Electronic Material Change (EMC) and an Electronic Material Bulletin (EMB) to cover the approved change. ECP's, EMC's and EMB's shall be prepared in accordance with SAR-300.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

Source: https://assist.dla.mil -- Downloaded: 20260-19T09:06Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Application - The equipment shall be suitable for testing and maintaining the above-named receiving sets in a workshop on shore or at sea.

6.2 The parentheses (*), in the type designation throughout this specification, shall be replaced by either a number or letter furnished by the Bureau of Aeronautics upon application by the contractor for assignment of nomenclature. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 20260-19T09:06Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:06Z
Check the source to verify that this is the current version before use.